

Jinmyeong Choi

1998.07.05

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SUMMARY

Master's student in Data Science at Seoul National University with a strong research focus on time series forecasting, representation learning, and foundation models. Adept at leading cross-functional teams, managing projects, and mentoring peers.

EDUCATION

Seoul National University, Seoul, Korea

2024.09 -

Master of Data Science

Seoul National University, Seoul, Korea

2018.03 - 2024.08

Bachelor of Arts in Economics, with a Minor in Data Science for Humanities

RESEARCH INTERESTS

- Time Series Forecasting
- Representation Learning
- Time Series Foundation Model

PUBLICATIONS

- **Choi, J.**, Shook, B. & Dubrawski, A.

Non-stationarity in the Embedding Space of Time Series Foundation Models (ICLR Workshop 2026)[[arxiv](#)].

- Investigated how different forms of non-stationarity—distributional shifts and temporal dependence violations—are preserved or suppressed in the embedding spaces of time series foundation models (TSFMs).
- Analyzed representation-level behavior under multiple metrics to provide a diagnostic perspective on how learned embeddings encode, entangle, or discard statistical properties of time series.

- **Choi, J.** & Dubrawski, A.

Towards Universal Representation-Based Process Control (Submitted to ICML 2026).

- Proposed a representation-based process control framework that tests whether a system operates consistently with an empirical reference regime using embeddings from time series foundation models.
- Developed a pipeline enabling consistency testing beyond classical stationarity tests (e.g., ADF, KPSS), allowing detection of regime deviations that traditional statistical tests cannot capture.

- Jang, J., Park, H., **Choi, J.**, & Kim, T.

Towards Robust Real-World Multivariate Time Series Forecasting: A Unified Framework for Dependency, Asynchrony, and Missingness (ICLR 2026)[[arxiv](#)].

- Introduced **ChannelTokenFormer**, a Transformer-based model that explicitly models cross-channel dependencies, handles asynchronous sampling, and manages missing values without relying on interpolation.
- Second author contribution: model architecture design and implementation, ablation studies, and empirical evaluation on benchmark and industrial datasets.

WORK EXPERIENCES

- **Remote Research Collaborator**, Auton Lab, Carnegie Mellon University 2026.02 – present
 - Continuing collaboration with Prof. Artur Dubrawski on representation learning and monitoring methods for time series foundation models
- **Research Intern**, Auton Lab, Carnegie Mellon University 2025.08 – 2026.02
 - Led an independent research project under the supervision of Prof. Artur Dubrawski investigating the internal representations of large-scale time series foundation models, including MOMENT and Chronos2.
 - Conducted systematic experiments to analyze how different forms of non-stationarity manifest in embedding space and developed representation-based methods for time series monitoring and process control.
- **Research Assistant for Professor Jeong Min Lee**, Seoul National University 2024.01 – 2024.02
 - Conducted data preprocessing and analysis for the evaluation of the "Ansim Income" policy using Python.
 - Collaborated with researcher from the Seoul Welfare Foundation to analyze datasets related to the policy's impact on welfare outcomes.

PROJECT EXPERIENCES

- **HR-legal assistance Platform** 2025.03 – 2025.06
 - Acted as the Product Manager and LLM Lead, directing the project roadmap, managing task distribution, and coordinating a cross-functional team to develop an AI-powered HR-legal assistance platform.
 - Architected and implemented an end-to-end Retrieval-Augmented Generation (RAG) pipeline that retrieves relevant documents from a database to generate precise, context-aware answers to complex legal and HR queries.
 - Engineered a human-in-the-loop feedback system for the iterative refinement of LLM-generated responses, continuously improving the accuracy and reliability of the platform.
- **Analysis of Consumer Goods as a Substitute for Price Index in Real Exchange Rate Measurement** 2022.12 – 2023.02
 - Investigated correlation between the real exchange rate and Big Mac Index, analyzing consumer goods as substitutes for real exchange rate measures
 - Received Excellence Prize from Computing Project Competition by Seoul National University Faculty of Liberal Education
- **Conversational AI – Term Project** 2024.09 – 2024.12
 - “From Rote Recital to Real Understanding: A Contextual-Semantic Approach to Humor Comprehension in LLMs”
 - Expanding humor datasets by rephrased and reversed variations.
 - Leading to improvements in both humor comprehension and logical reasoning.

EXTRACURRICULAR ACTIVITY

- **Python TA**, Seoul National University 2026.03 – present
 - Lead hands-on lab sessions for a Python programming course, teaching fundamental programming concepts and guiding students through practical coding exercises.
- **Python Tutor**, Seoul National University 2023.03 – 2024.12
 - Tutored students in a Python basics course, focusing on fundamental programming concepts and syntax.
- **DTM Supporters**, Seoul National University 2021.09 – 2021.12
 - Assisted international students with administrative tasks and facilitated cultural adaptation.
- **Leadership Roles:**
 - President, 7th L&K Sports Business Forum 2022.03 – 2022.08
 - Seminar Team Leader, 1st Block Leader, Seoul National University 32nd Model United Nations

CLUB & VOLUNTEER ACTIVITIES

- Member, SNU Go (Baduk) Club *2023.09 – present*
- Member, SNU Sunshine Volunteer *2021.09 – 2022.02*
- Member, SNU Model United Nations *2018.09 – 2019.12*

LANGUAGES

- English – High Proficiency (TEPS 465, TOEFL 100 $\approx$ CEFR C1)
- Mandarin - Intermediate

SKILLS

- Programming Languages: Python (Advanced), C++ (Intermediate), C (Intermediate)
- AI/ML Libraries: PyTorch(Advanced), Pandas (Advanced), Scikit-learn, Matplotlib

OTHERS

- Republic of Korea Army, Sergeant, Honorably Discharged (*2020.01 – 2021.07*)
 - Served as Squad Leader, responsible for leading and managing a unit of soldiers in training and operations.
 - Completed Airborne/Paratrooper training, demonstrating physical endurance, adaptability, and teamwork under high-pressure conditions.